

Streaming Camera Setup and Pricing

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This document summarizes the full setup for the camera streaming system and gives realistic budget ranges you can share with a client.

What the system does

The system takes a camera that already outputs RTSP, converts that stream into browser-friendly HLS, and makes it available in a normal web browser.

Live path

- 1 Camera -> RTSP
- 1 FFmpeg -> HLS
- 1 Node server or Vercel app -> HTTPS
- 1 Cloudflare Tunnel or public web app -> browser

Browsers do not play RTSP directly, so FFmpeg repackages the camera stream into HLS. The tradeoff is a small delay, typically about 6-10 seconds behind real time.

Two deployment options

Option A: Local always-on device + Cloudflare Tunnel

This is the simplest setup.

How it works:

- 1 The camera stays on-site.
- 1 A small computer on the same network runs the server.
- 1 FFmpeg pulls the RTSP feed and writes HLS segments.
- 1 Cloudflare Tunnel publishes the local server to the internet.

Best for:

- 1 Lowest cost
- 1 Simple setup
- 1 No router port forwarding
- 1 No cloud upload workflow

Option B: Camera-side agent + Vercel-hosted viewer

This version is useful when you want the viewer app hosted in the cloud.

How it works:

- 1 A machine on the camera network runs the agent.
- 1 The agent pulls RTSP and uploads live HLS segments to Vercel Blob.
- 1 The Vercel app serves the viewer and proxies the live segments.

Best for:

- 1 Hosting the viewing app in the cloud
- 1 Not depending on a local tunnel for viewer access
- 1 Cleaner access from anywhere, while still keeping the camera on-site

Hardware needed

Required

- 1 IP camera with RTSP support
- 1 Always-on device on the same network as the camera
- 1 Power supply for that device
- 1 Internet connection

Recommended always-on device choices

1. Raspberry Pi 5

Estimated one-time cost:

- 1 Raspberry Pi 5: \$80-\$120
- 1 Power supply: \$10-\$15
- 1 Case and cooling: \$10-\$25
- 1 microSD card: \$10-\$25
- 1 Optional SSD setup: \$40-\$80 extra

Estimated total:

- 1 Basic setup: \$110-\$185
- 1 More reliable SSD setup: \$150-\$265

Good for:

- 1 H.264 passthrough
- 1 Low-power always-on use
- 1 Small, budget-friendly installs

Limitations:

- 1 Less comfortable if transcoding is needed
- 1 microSD is less durable than SSD for 24/7 use

2. Intel N100 mini PC

Estimated one-time cost:

- 1 Mini PC: \$150-\$300
- 1 Optional SSD upgrade if needed: usually included, otherwise \$30-\$80

Estimated total:

- 1 \$150-\$300

Good for:

- 1 The best balance of cost and reliability
- 1 Better headroom for FFmpeg
- 1 Safer choice if the camera may need transcoding

This is usually the best recommendation if the client wants something dependable and low-maintenance.

3. Used small PC or thin client

Estimated one-time cost:

- 1 \$50-\$200 used

Good for:

- 1 Lowest hardware cost if one is already available
- 1 Basic always-on streaming duties

Tradeoffs:

- 1 Higher power draw than a Pi or mini PC in some cases
- 1 Quality and reliability depend on the specific machine

4. UPS battery backup, optional

Estimated cost:

- 1 \$60-\$150

Why it matters:

- 1 Keeps the stream online during short power interruptions
- 1 Protects the device from unexpected shutdowns

Software and recurring costs

Local tunnel setup

- 1 Cloudflare quick tunnel: \$0
- 1 Cloudflare account: \$0
- 1 Custom domain for a stable public URL: usually \$10-\$20 per year
- 1 Named Cloudflare tunnel: \$0 extra beyond the domain

Cloud-hosted viewer setup

- 1 Vercel app: plan-dependent
- 1 Vercel Blob storage and bandwidth: usage-dependent
- 1 Agent software: no license fee

For a client quote, it is safest to treat the cloud-hosting cost as variable and tied to usage, plan choice, and bandwidth.

Other recurring costs

- 1 Electricity for the always-on device
- 1 Internet service at the camera site
- 1 Domain renewal, if a custom URL is wanted

Budget ranges

Lowest-cost working setup

- 1 Raspberry Pi 5 bundle
- 1 microSD card
- 1 Cloudflare quick tunnel

Estimated total: \$110-\$185

Recommended reliable setup

- 1 Intel N100 mini PC
- 1 Cloudflare quick tunnel or named tunnel
- 1 Optional SSD included in the mini PC

Estimated total: \$150-\$300

More robust setup

- 1 Intel N100 mini PC
- 1 SSD-based storage
- 1 UPS battery backup
- 1 Custom domain
- 1 Named Cloudflare tunnel

Estimated total: \$220-\$450

What the client needs to provide

- 1 Camera RTSP URL
- 1 Camera username and password, if required
- 1 A place to keep the always-on device near the camera
- 1 Network access to the camera
- 1 If using a custom public URL, a domain name

Setup steps

Local tunnel version

- 1 Copy the environment file and set the camera RTSP URL.
- 2 Install dependencies.
- 3 Verify the camera feed with the probe command.
- 4 Start the local server.
- 5 Start the tunnel.
- 6 Open the public URL.

Cloud-hosted version

- 1 Set up the Vercel project.
- 2 Add the Blob token to the agent environment.
- 3 Point the agent at the camera RTSP URL.
- 4 Start the agent on the always-on device.

- 5 Open the Vercel app.

Operational notes

- 1 The live feed usually lags the camera by 6-10 seconds.
- 1 The system is designed for browser playback, not ultra-low-latency viewing.
- 1 If the camera outputs H.265/HEVC, transcoding may be required.
- 1 The camera-side machine must stay on for the feed to remain available.
- 1 Password protection can be enabled for shared links.

Recommended client summary

If the client wants the simplest and most affordable setup, the best recommendation is:

- 1 one small always-on device on-site
- 1 Cloudflare Tunnel for public access
- 1 optional password protection
- 1 optional UPS if uptime matters

If the client wants a stronger all-around option, recommend an Intel N100 mini PC instead of a Raspberry Pi.